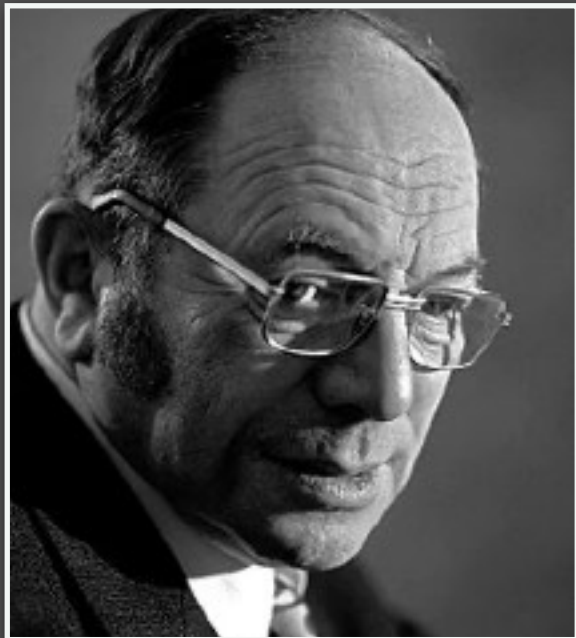




Kantorovich's Problem:

(Continuous version dual)

$$\sup_{f,g : f+g \leq c} \int_X f \, d\alpha + \int_Y g \, d\beta$$



Kantorovich's Problem:

(Continuous version dual w/ c -conjugacy)

$$\sup_{f \in c\text{-CVX}} \int_X f \, d\alpha + \int_Y f^c \, d\beta$$

One dual variable instead of two need efficient way to compute conjugate

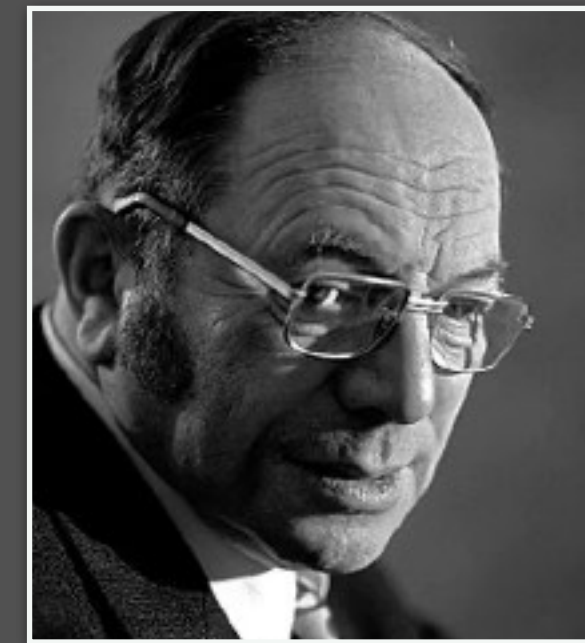
A reformulated dual



Kantorovich's Problem:

(Continuous version dual)

$$\sup_{f, g : f+g \leq c} \int_X f \, d\alpha + \int_Y g \, d\beta$$



Kantorovich's Problem:

(Continuous version dual w/ c -conjugacy)

$$\sup_{f \in c\text{-CVX}} \int_X f \, d\alpha + \int_Y f^c \, d\beta$$

What did we gain?

One dual variable instead of two, but **need efficient way to compute c -conjugate**

Particular cases:

- $c(x, y) = \|x - y\|$ (1-Wasserstein): c -convexity = 1-Lipschitz, $f^c = -f$
- $c(x, y) = \|x - y\|^2$ (2-Wasserstein): c -convexity = classic convexity, $f^c =$ convex conjugate, aka Legendre-Fenchel transform; $f^*(y) = \sup_x \langle x, y \rangle - f(x)$

Section 2: Deriving the Dual